Steady Heat Conduction in a 2D Plate

Solution of Laplace Equation Using Fourier Series

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Physical Problem

We study steady heat conduction in a rectangular plate.

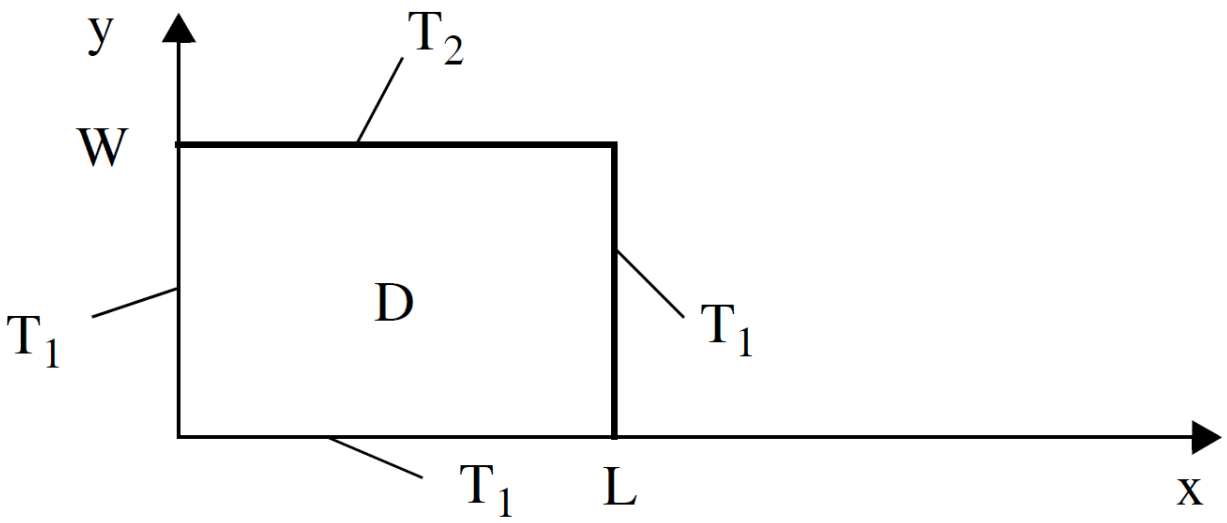
- No internal heat generation
- Steady state
- Constant thermal conductivity

Governing equation:

$$\nabla^2 T = 0$$

This is the **Laplace equation**.

Geometry and Boundary Conditions



Boundary conditions:

$$T(0, y) = 0$$
$$T(L, y) = 0$$
$$T(x, 0) = 0$$
$$T(x, W) = 1$$

Laplace Equation

For steady conduction:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

We solve using:

Method

Separation of variables

Assume:

$$T(x, y) = X(x)Y(y)$$

Separation of Variables

Substituting:

$$X''Y + XY'' = 0$$

Divide by XY:

$$\frac{X''}{X} = -\frac{Y''}{Y} = \lambda$$

We obtain two ODEs:

$$X'' + \lambda X = 0$$

$$Y'' - \lambda Y = 0$$

Solution in x Direction

Boundary conditions:

$$T(0, y) = T(L, y) = 0$$

Thus:

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right)$$

Eigenvalues:

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2$$

Solution in y Direction

We solve:

$$Y'' - \left(\frac{n\pi}{L}\right)^2 Y = 0$$

General solution:

$$Y_n(y) = A_n e^{n\pi y/L} + B_n e^{-n\pi y/L}$$

Using hyperbolic functions:

$$Y_n(y) = C_n \sinh\left(\frac{n\pi y}{L}\right)$$

because:

$$T(x, 0) = 0$$

General Solution

Combining solutions:

$$T(x,y) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right) \sinh\left(\frac{n\pi y}{L}\right)$$

Unknown coefficients C_n come from the last boundary condition.

Applying the Top Boundary

Boundary:

$$T(x, W) = 1$$

Therefore:

$$1 = \sum_{n=1}^{\infty} C_n \sinh\left(\frac{n\pi W}{L}\right) \sin\left(\frac{n\pi x}{L}\right)$$

We must expand the constant function using a Fourier sine series.

Fourier Sine Expansion

We express:

$$1 = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right)$$

Coefficient formula:

$$A_n = \frac{2}{L}\int_0^L \sin\left(\frac{n\pi x}{L}\right)\, dx$$

Computing the Integral

$$A_n = \frac{2}{L}\left[-\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\right]_0^L$$

$$A_n = \frac{2}{n\pi}\left(1-(-1)^n\right)$$

Important result:

- Even $n \rightarrow$ coefficient $= 0$
- Only odd modes survive

Determining C_n

Comparing both expansions:

$$A_n = C_n \sinh\left(\frac{n\pi W}{L}\right)$$

Thus:

$$C_n = \frac{2(1-(-1)^n)}{n\pi \sinh(n\pi W/L)}$$

Final Temperature Distribution

$$T(x,y) = \sum_{n=1}^{\infty} \frac{2(1-(-1)^n)}{n\pi \sinh(n\pi W/L)} \sin\left(\frac{n\pi x}{L}\right) \sinh\left(\frac{n\pi y}{L}\right)$$

This is the analytical solution of the 2D steady heat conduction problem.

Physical Interpretation

- Each term represents a thermal mode
- Higher modes decay faster inside the plate
- Heat diffuses smoothly from the hot boundary
- Solution satisfies all boundary conditions simultaneously

Connection to Finite Elements

Why is this important?

- Analytical benchmark solution
- Used to validate FEM implementations
- Shows eigenfunction basis idea
- FEM approximates this infinite expansion numerically

Summary

1. Physical model $\rightarrow$ Laplace equation
2. Separation of variables
3. Eigenvalue problem
4. Hyperbolic solutions
5. Fourier expansion
6. Final analytical solution